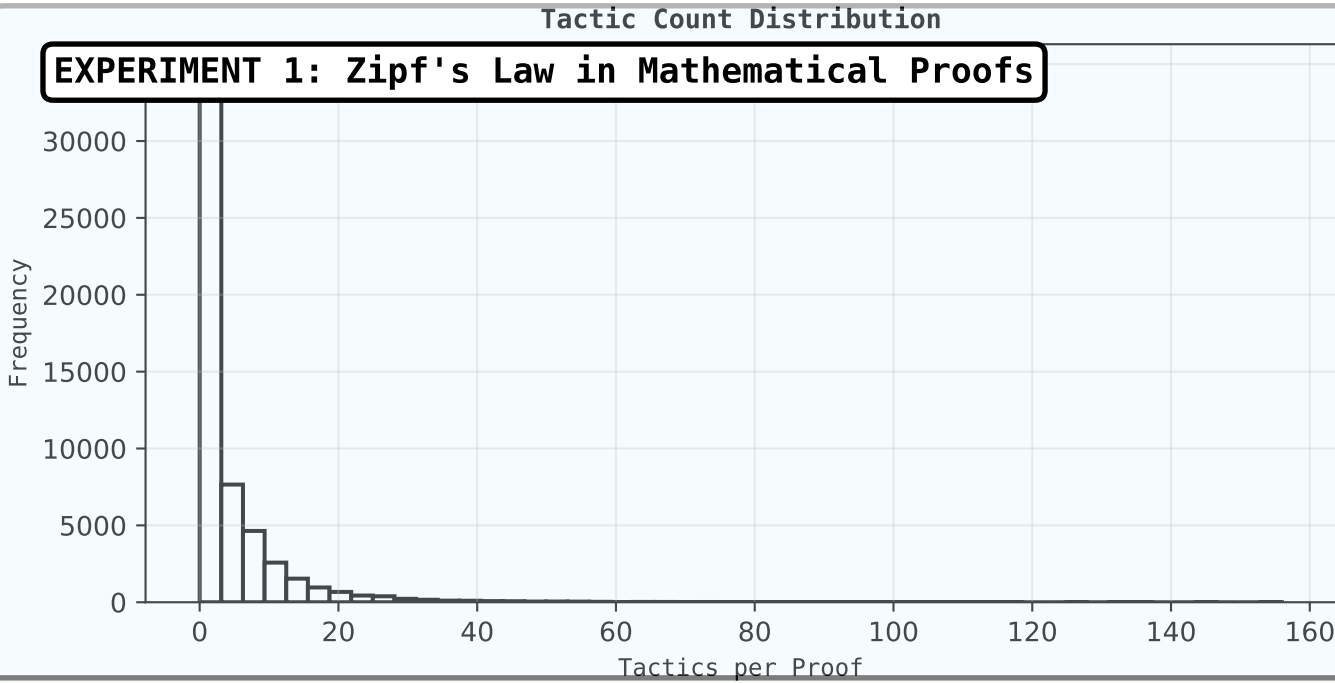
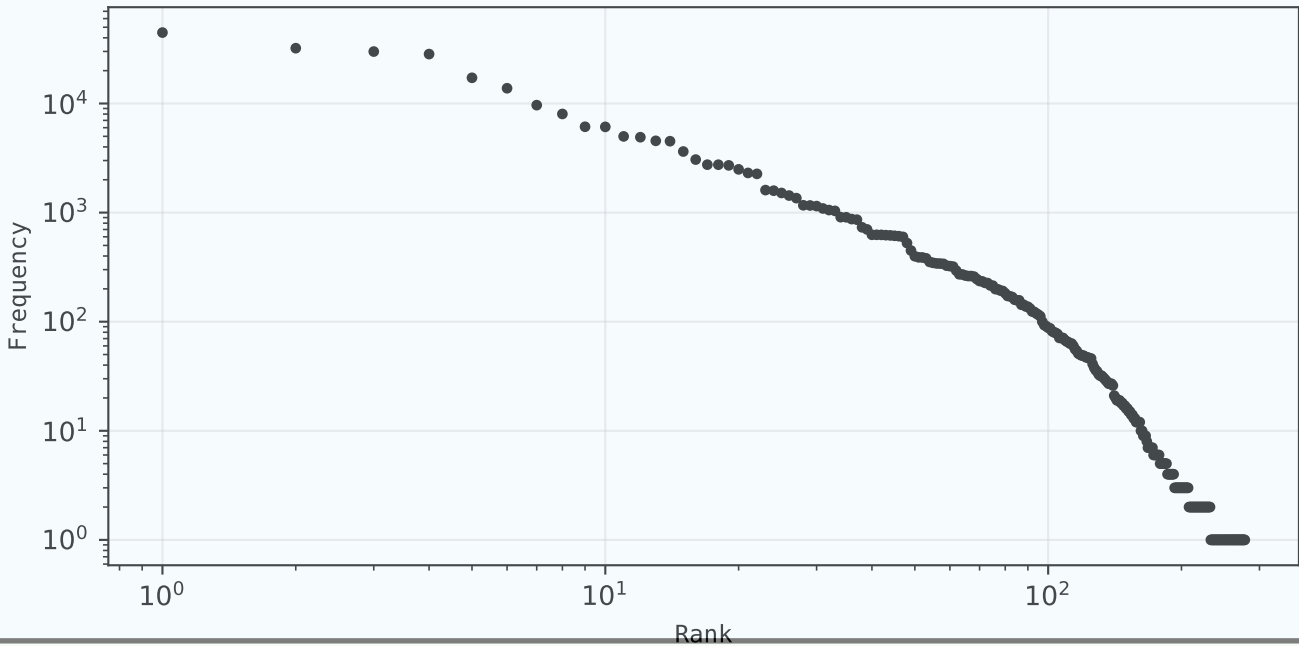


Mathlib Description Length: Complete Analysis with MDL Gain

EXPERIMENT 1: Zipf's Law in Mathematical Proofs



Zipf's Law (log-log)

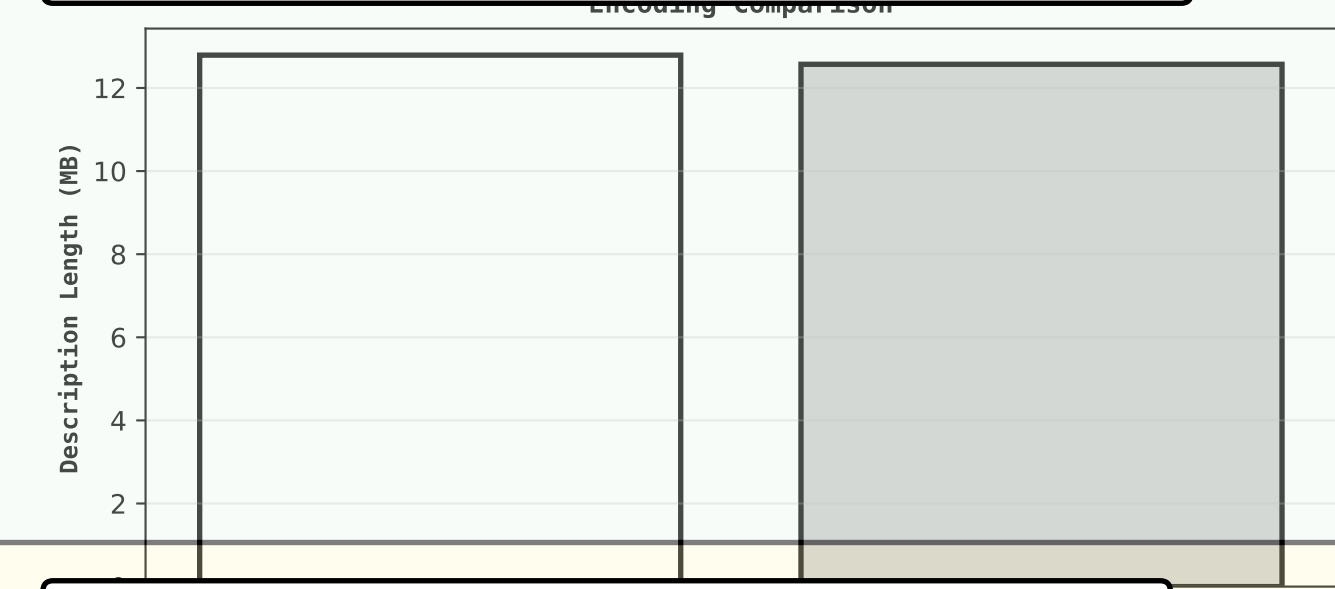


Finding: Tactic frequencies follow power-law distribution (Zipf's law)

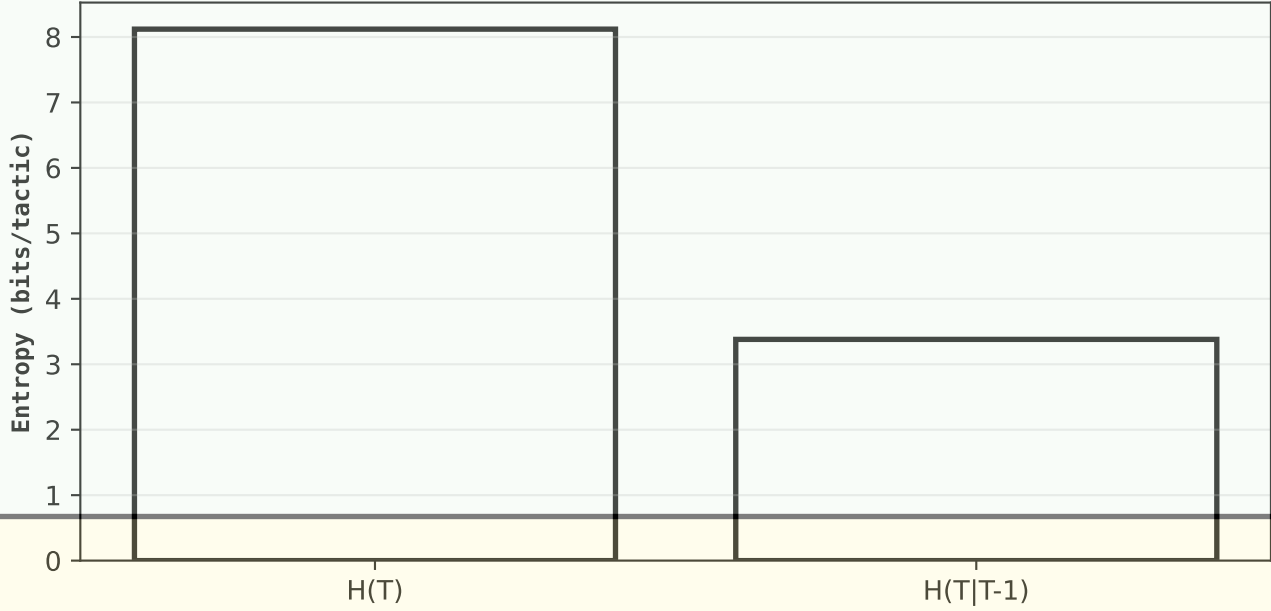
Most proofs are short (median ~2 tactics), but tail extends to 156.

Top tactics: rw, simp, exact, · (bullet point for proof structure)

EXPERIMENT 2: Frequency-Based Compression (1.02x gain)



Tactic Predictability: 58.4%

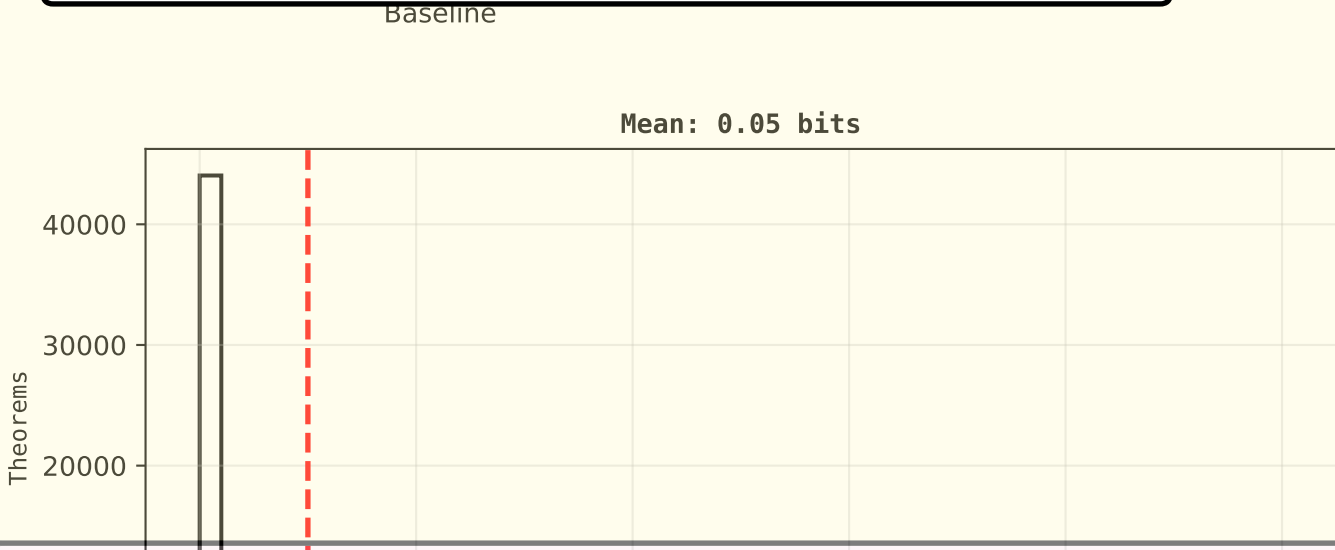


Finding: Shannon encoding achieves only 1.02x compression.

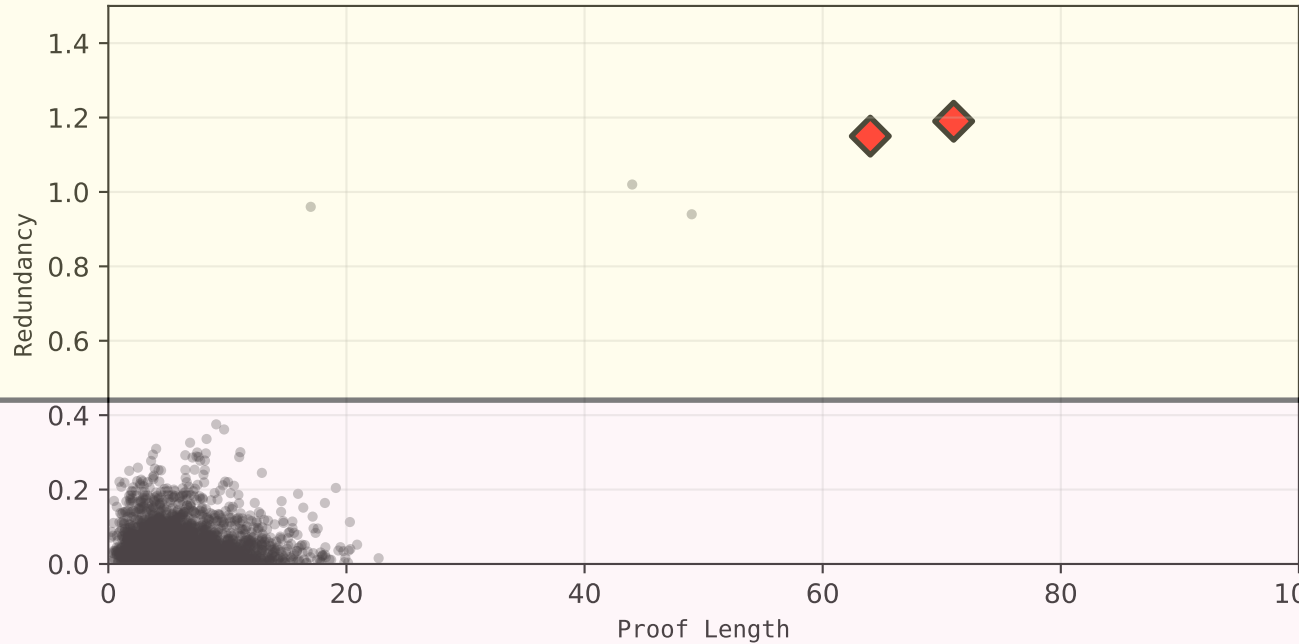
Human factorization already captures most frequency-based optimization.

58.4% of tactics predictable from previous tactic (H reduces 8.12~3.38).

EXPERIMENT 3: Per-Theorem Compression (99.8% optimal)



Outliers: 36x/40x "have"

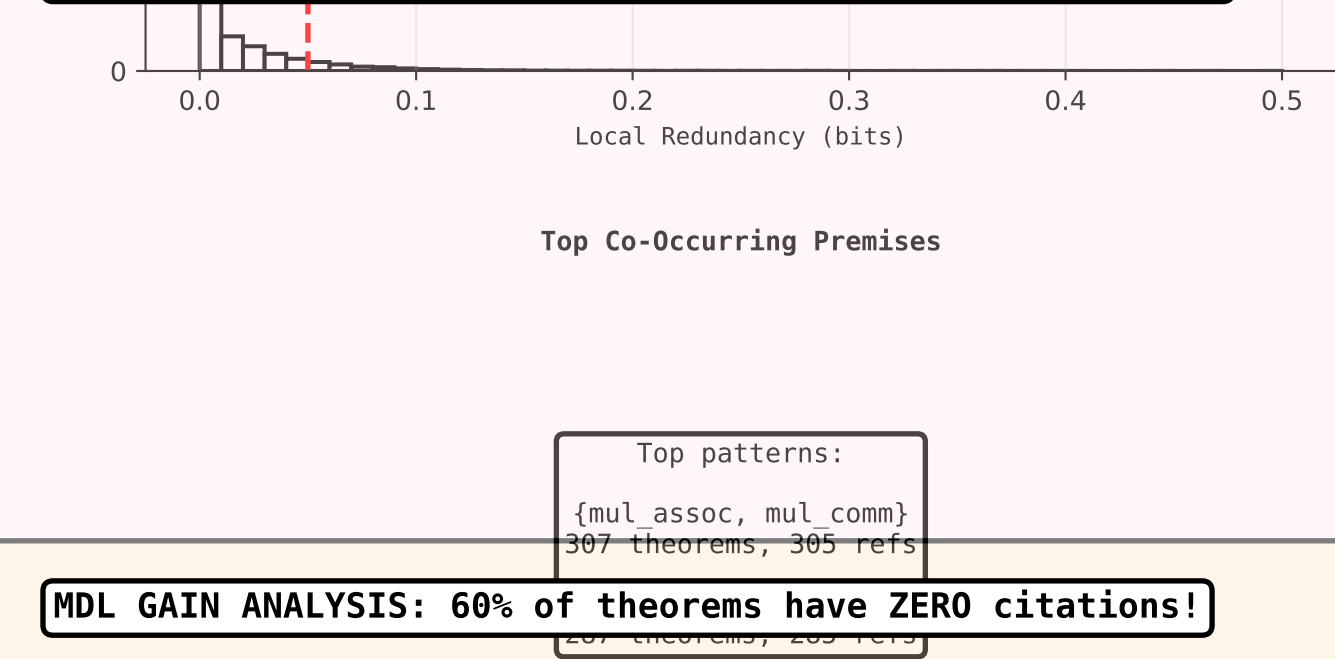


Finding: Average local redundancy is only 0.05 bits per theorem.

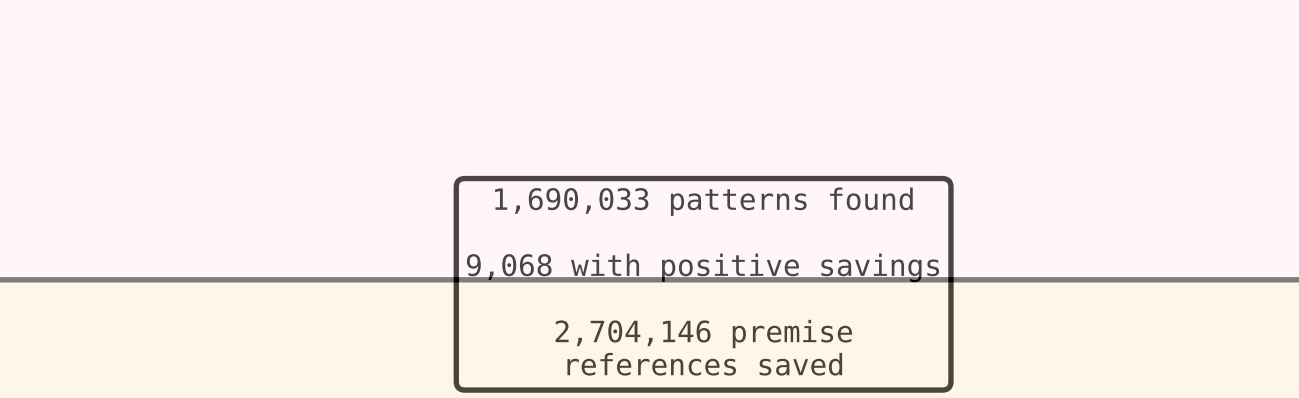
99.8% already optimally factored. Only ~100 outliers (0.2%) show repetitive patterns (have chains).

These are deliberate style choices for clarity, not missed abstractions.

CRYSTALLIZATION: Premise Co-Occurrence (2.7M refs saved)



Crystallization Statistics



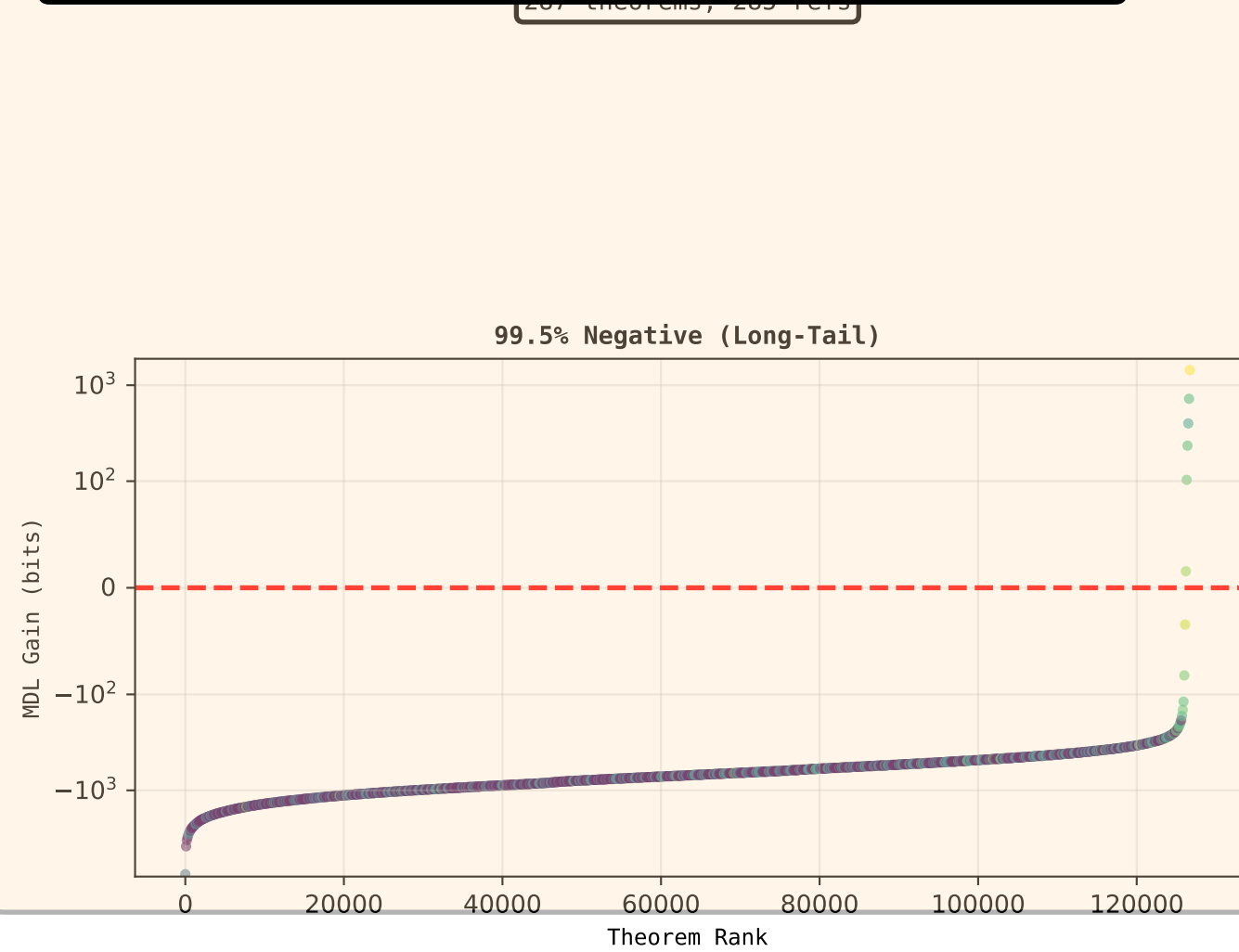
Finding: Premise co-occurrence reveals 33x more compression potential than tactic patterns!

Fundamental combinations:

- Multiplication laws (assoc+comm)
- Sum constructors (inl+inr)
- Category theory (comp+id)

True crystallization measures WHAT theorems prove (content), not HOW they're proven (style).

MDL GAIN ANALYSIS: 60% of theorems have ZERO citations!

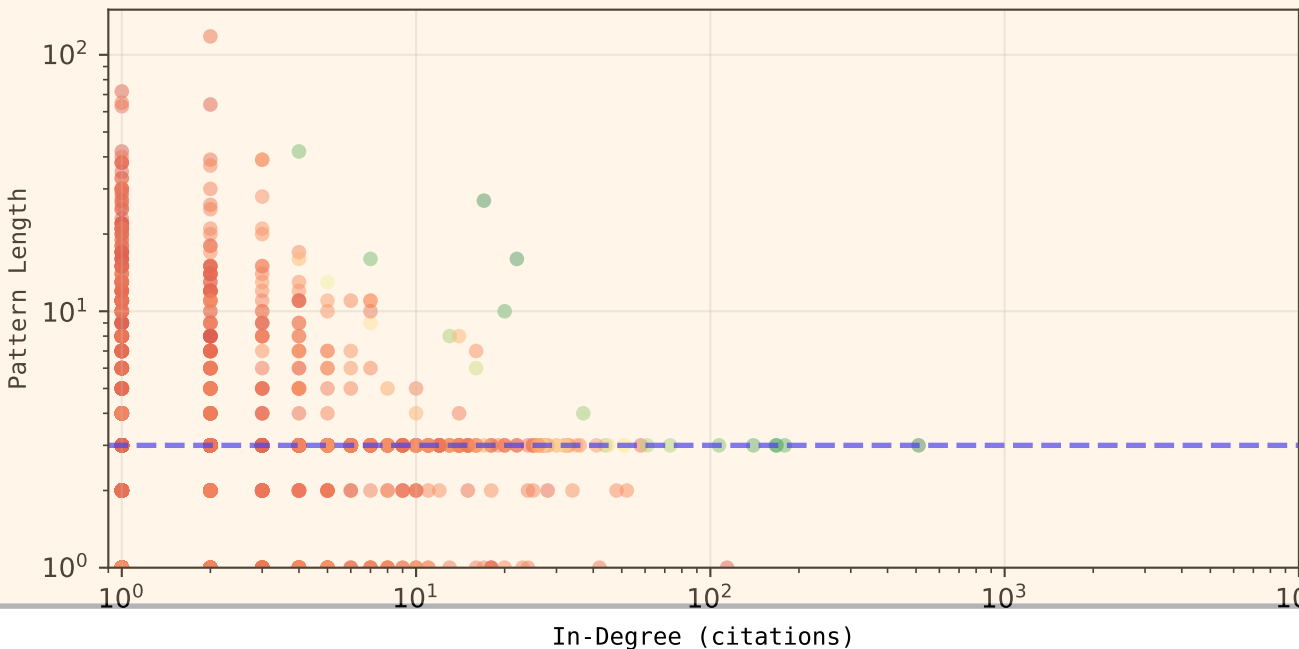


1,690,033 patterns found

9,068 with positive savings

2,704,146 premise references saved

Phase Space



Finding: Long-tail distribution!

- 60% have 0 citations
- 80% have ≤1 citation
- Only 3% have 10+ citations
- Only 0.2% have 100+ citations

Top compressive theorems:

- trans (11K bits, 1538 uses)
- mul_comm (9.6K, 1288 uses)
- mul_assoc (7.7K, 1056 uses)

Interpretation: Most theorems are specialized results. Few fundamental lemmas carry the compression load.

Negative MDL ≠ useless! Many provide:

- Conceptual organization
- Implicit uses (type classes)
- Stepping stones for proofs